

Hawkeye EQMS · CAPA v2 — Corrective & Preventive Action · Feature Guide

Pharma-EQMS comparison · click-by-click guide · personas + lifecycle walkthrough · AI map · test results.

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1. Module overview

Module	CAPA v2 — Corrective & Preventive Action
Pharma purpose	Formalised corrective + preventive action lifecycle. Intake → triage → investigation → RCA → action plan → implementation → effectiveness review → closure. 5 mandatory approval gates.
Compliance basis	ISO 9001:2015 §10.2 · 21 CFR 211.192 · 21 CFR 820.100 (CAPA · medical device) · ICH Q10 §3.2.2 · EU GMP Chapter 1.4
Hawkeye routes	/buyer/capas, /auditor/capas (frontend) · /api/capa-v2/* (backend)
Hawkeye model file	backend/src/models/capaV2Models.js (17 collections)
Tenant module flag	modules.CAPA

CAPA v2 supersedes the legacy single-collection capas with a 17-collection event-sourced model: every state transition writes a capa-v2-status-history row, every approval writes a capa-v2-approvals row, every action item is its own document, and every implementation artefact has its own capa-v2-implementation-evidence row. This makes it possible to satisfy 21 CFR Part 11 audit-trail requirements without separate logging — the data IS the audit trail. Every state transition is gated by the statusMachine.js state graph (no skipping states without admin override) and by role-based permits.

2. Standard pharma EQMS vs Hawkeye — gap analysis

Each row is a capability a pharma QA team would expect from an EQMS deviation module. Hawkeye column shows what's actually shipped. **MET** = parity, **PARTIAL** = ships but with caveats, **GAP** = not implemented yet.

Pharma expectation	Hawkeye implementation	Status	Note
5-stage approval workflow (Triage / RCA / Action Plan / Effectiveness / Closure) 21 CFR 820.100	POST /api/capa-v2/capas/:id/approvals with stage enum + decision (APPROVED/REJECTED/NEEDS_REWORK). Per-stage gating in statusMachine.js.	MET	

Pharma expectation	Hawkeye implementation	Status	Note
Auto-generated CAPA number (CAPA-YYYY-NNNN) GMP traceability	nextCapaNumber() in modules/capaV2/prefillService.js. Per-tenant sequence with year prefix.	MET	
Multiple sources: audit finding, deviation, complaint, regulator inspection ICH Q10 §3.2.5	sourceType enum on intake + capa records: AUDIT_FINDING / DEVIATION / COMPLAINT / INSPECTION / INTERNAL / CUSTOMER_FEEDBACK. capa-v2-source-links table for traceability.	MET	
Triage decision: Formal CAPA / Correction Only / No CAPA Needed ICH Q10 §3.2.2	POST /triage/:id/decision with decision enum + rationale. Drives next state.	MET	
Structured RCA capture (5-Why + Fishbone + FMEA + verification) ICH Q9	PUT /capas/:id/root-cause with rcaMethod enum + rootCauseStatement + verification. Stored in capa-v2-root-causes.	MET	
Action plan with corrective + preventive + interim actions, each with owner + due date + status 21 CFR 820.100(a)	PUT /capas/:id/action-plan + POST /capas/:id/action-items. Each action-item is its own document with status NOT_STARTED → IN_PROGRESS → COMPLETED.	MET	
Implementation evidence (uploaded documents, test results) 21 CFR 211.192	POST /capas/:id/implementation-evidence — accepts S3 refs + metadata. Stored in capa-v2-implementation-evidence.	MET	
Effectiveness check after implementation (PASS / FAIL) 21 CFR 820.100(a)(4)	PUT /capas/:id/effectiveness with result enum + reviewWindow. Stored in capa-v2-effectiveness-checks.	MET	
Closure: CLOSED_EFFECTIVE or CLOSED_INEFFECTIVE 21 CFR 820.100(a)(5)	POST /capas/:id/close with outcome (EFFECTIVE / INEFFECTIVE) + closureReason. Final approval gate.	MET	
Re-open if recurrence detected ICH Q10 §3.2.4	POST /capas/:id/reopen with reason. Status flips to REOPENED.	MET	
Risk assessment captured (patient / product / compliance / recurrence) ICH Q9	POST /capas/:id/risk-assessment. Stored in capa-v2-risk-assessments.	MET	
Audit trail of every state transition (ALCOA+) 21 CFR Part 11 §11.10(e)	Every transitionCapaStatus() call writes a capa-v2-status-history row with from/to/userId/reason/timestamp. Mongoose pre-save hook.	MET	
Daily metric snapshot (age, overdue, recurrence) QMS KPI	createMetricSnapshot() called on close + reopen. capa-v2-metric-snapshots collection.	MET	
Similar / recurrence / duplicate detection ICH Q10 §3.2.4	capa-v2-similarity-links table with linkType (SIMILAR / RECURRENCE / POTENTIAL_DUPLICATE). Population currently manual.	PARTIAL	AI-driven similarity scoring on roadmap
Cross-module CAPA creation from deviation/complaint/finding via single button —	POST /api/quality/capa-from-deviation creates a real v2 record + action plan + links. Wired to AI Draft RCA drawer.	MET	
AI: predictive on-time / effectiveness scoring FDA AI guidance Jan 2025	POST /api/ai/predict/capa-outcome (Wave 3 heuristic). Returns P(on-time) + P(effective) + top factors.	MET	

3. Personas + role mapping

Login password for every persona: `EqmsDemo@2026`. Roles drive which API endpoints respond — see Hawkeye `roleMiddleware.permit()`.

Persona	Email	Responsibilities in this module	Lifecycle states they touch
Audit Lead Maria auditor	audit.lead@novex-pharma.demo	Reviews intake, makes triage decision, approves RCA + action plan + closure. Owns the approval gates.	UNDER_TRIAGE RCA_PENDING_APPROVAL ACTION_PLAN_PENDING_APPROVAL EFFECTIVENESS_REVIEW CLOSED
QA Specialist Kenji admin	qa.specialist@novex-pharma.demo	Submits intakes from deviations / complaints, executes investigation + RCA + action items, uploads evidence.	INTAKE_DRAFT INVESTIGATION_IN_PROGRESS IN_IMPLEMENTATION AWAITING_EFFECTIVENESS_CHECK
Head of QA James admin	qa.head@novex-pharma.demo	Owns CAPAs end-to-end across the tenant; assigns ownership; signs off effectiveness.	CAPA_OPEN onwards
VP Quality Elena tenant_admin	vp.quality@novex-pharma.demo	Final closure approval on CRITICAL CAPAs; reviews CAPA backlog at MRM.	CLOSED_EFFECTIVE CLOSED_INEFFECTIVE

4. Feature catalogue (click-by-click)

4.1 CAPA workspace dashboard

What it does	Counts of open / closed-effective / closed-ineffective / overdue CAPAs + bySeverity breakdown + overdue action-items count.
Menu / route	/buyer/capas (admin/buyer view) or /auditor/capas (auditor view)
Available to roles	admin buyer auditor tenant_admin
Backend API	GET /api/capa-v2/dashboard

CLICK-BY-CLICK

- NAVIGATE

 Click 'CAPAs' in the top nav
→ Dashboard renders with counter cards
- WAIT

 Spinner clears (~700 ms)
→ counts.open · counts.closedEffective · counts.overdue · bySeverity render

Audits

New Audit

CAPAs

Documents

Deviations

Complaints

Changes

Training

Equipment

Risk

Reviews

Buyer > CAPAs

Admin

CAPA Management Workspace

OPEN
5

OVERDUE
0

RECURRING
0

NEW CANDIDATES
0

Audit ID

Generate

Manual CAPA title

Create

Search candidates

Candidate Status
All

You don't have any capa candidate queue

Search CAPAs

CAPA Status
All

Refresh

Q Fuzzy search

Density
Standard

View

Save As

Save View

Reset

Manage

Export to CSV

Manage Columns

CAPA ID	Title	Status	Severity	Due Date	Owner
CAPA-2026-00005	CAPA from DEV-2026-0064: F1 X-mod · OOS dissolu...	CAPA_OPEN	HIGH	5/25/2026, 11:05:21 ...	supplier_quality_lead
CAPA-2026-00004	CAPA from DEV-2026-0063: F1 X-mod · OOS dissolu...	CAPA_OPEN	HIGH	5/25/2026, 11:02:31 ...	supplier_quality_lead
CAPA-2026-00003	CAPA from DEV-2026-0062: F1 X-mod · OOS dissolu...	CAPA_OPEN	HIGH	5/25/2026, 10:59:15 ...	supplier_quality_lead
CAPA-2026-00002	CAPA from DEV-2026-0061: F1 X-mod · OOS dissolu...	CAPA_OPEN	HIGH	5/25/2026, 10:54:59 ...	supplier_quality_lead
CAPA-2026-00001	CAPA from DEV-2026-0060: P	CAPA_OPEN	HIGH	5/25/2026, 10:54:13 ...	supplier_quality_lead

CAPA Detail

Overview

Investigation

Action Plan

Effectiveness

History

CAPA-2026-00005 | CAPA_OPEN

CAPA from DEV-2026-0064: F1 X-mod · OOS dissolution on demo batch

Owner User ID

Owner Role

69e64e7a69b2ba745d40bbf

supplier_quality_lead

Assign

Stage

Decision

Approval Note

Approve

RCA

APPROVED

Comment

Add Comment

Close Effective

Reopen

state-screens/capa-buyer.png

4.2 Candidate from audit finding

What it does	Promote an audit finding to a CAPA candidate so it's tracked in the workspace before formal intake.
Menu / route	Audit detail · finding row · 'Promote to CAPA candidate' button (or POST API)
Available to roles	auditoradmintenant_admin
Backend API	POST /api/capa-v2/candidates/from-finding

CLICK-BY-CLICK

- CLICK

On an audit finding, click 'Promote to CAPA candidate'
→ capa-v2-candidates row created with status NEW; sourceType=AUDIT_FINDING

4.3 Bulk-create candidates from audit

What it does	One-click — every Major+ finding from a closed audit becomes a CAPA candidate.
Menu / route	Audit closure step · 'Generate CAPAs from findings' button
Available to roles	auditortenant_admin

Backend API

POST /api/capa-v2/candidates/bulk-from-audit/:auditId

CLICK-BY-CLICK

1. **CLICK** On the audit closure page, click 'Generate CAPAs'
→ One candidate per finding above the configured severity threshold

4.4 Create + submit intake

What it does

Author the intake form (issueStatement, scope, regulatoryExposure). Submit moves the candidate into UNDER_TRIAGE.

Menu / route

/buyer/capas → 'New CAPA intake' drawer

Available to roles

admin

buyer

auditor

Backend API

POST /api/capa-v2/intakes · POST /api/capa-v2/intakes/:id/submit

CLICK-BY-CLICK

1. **CLICK** Click 'New CAPA intake'
→ Drawer opens
2. **TYPE** Fill issueStatement (1 sentence)
→ required
3. **TYPE** Fill issueDescription (full narrative)
→ required
4. **CLICK** Choose severity (CRITICAL / MAJOR / MINOR)
→ default MINOR
5. **CLICK** Click 'Save' (status=DRAFT)
→ Intake row appears with state INTAKE_DRAFT
6. **CLICK** Click 'Submit for triage'
→ Status flips to UNDER_TRIAGE; triage row created

FIELDS CAPTURED

Field	Required	Validation / values	Notes
issueStatement	Yes	string	
issueDescription	Yes	multi-line text	
severity	No	CRITICAL MAJOR MINOR	default MINOR
sourceType	No	AUDIT_FINDING DEVIATION COMPLAINT INTERNAL INSPECTION	auto-set when promoted from a source record

4.5 Triage decision

What it does

Triage reviewer (auditor) decides FORMAL_CAPA / CORRECTION_ONLY / NO_CAPA_NEEDED.

Menu / route

Triage row · 'Triage decision' button

Available to roles

auditor

tenant_admin

admin

superadmin

Backend API

POST /api/capa-v2/triage/:id/decision

CLICK-BY-CLICK

1. **CLICK** Click 'Triage decision'
→ Drawer opens
2. **CLICK** Choose decision: FORMAL_CAPA_REQUIRED / CORRECTION_ONLY / NO_CAPA_NEEDED
→ Drives the next state
3. **TYPE** Fill rationale
→ required
4. **CLICK** Click 'Submit decision'

→ If FORMAL_CAPA → CAPA_OPEN; if CORRECTION_ONLY → IN_IMPLEMENTATION; if NO_CAPA → TRIAGE_NO_CAPA terminal

4.6 Open CAPA + assign owner

What it does	Mint the CAPA-v2 record once triage decides FORMAL_CAPA. Owner + due date required.
Menu / route	Triage decision flow · 'Create CAPA' button
Available to roles	triageRoles
Backend API	POST /api/capa-v2/capas

CLICK-BY-CLICK

1. [CLICK](#) Click 'Create CAPA' on the triage drawer
→ Drawer expands
2. [CLICK](#) Pick owner from user dropdown
→ ownerId required
3. [CLICK](#) Set dueDate (default +30 days)
→ required
4. [CLICK](#) Click 'Open CAPA'
→ Status=CAPA_OPEN; capaNumber generated; auto-redirect to detail

4.7 Save investigation

What it does	Investigator records investigation scope + summary + findings.
Menu / route	CAPA detail · Investigation tab
Available to roles	manageRoles
Backend API	PUT /api/capa-v2/capas/:id/investigation

CLICK-BY-CLICK

1. [CLICK](#) Open Investigation tab
→ Form renders
2. [TYPE](#) Fill investigationScope, summary, findings, evidenceLinks[]
→ All optional but recommended
3. [CLICK](#) Click 'Save investigation'
→ Status moves to INVESTIGATION_IN_PROGRESS

4.8 Save root-cause analysis

What it does	Capture the structured RCA — method enum + statement + verification.
Menu / route	CAPA detail · RCA tab
Available to roles	manageRoles
Backend API	PUT /api/capa-v2/capas/:id/root-cause
AI assistance	AI AI 5-Why scaffolder (Wave 1 · POST /api/ai/deviation/scaffold-five-why) can pre-populate the 5 whys

CLICK-BY-CLICK

1. [CLICK](#) Open RCA tab
→ Form renders
2. [CLICK](#) Choose rcaMethod (FIVE_WHY / FISHBONE / FMEA / FAULT_TREE / PARETO)
→ required
3. [TYPE](#) Fill rootCauseStatement
→ required
4. [CLICK](#) Click 'Submit for approval'

→ Status moves to RCA_PENDING_APPROVAL

4.9 Approve stage (RCA / Action Plan / Effectiveness / Closure)

What it does	Approver reviews the stage submission and APPROVES / REJECTS / NEEDS_REWORK.
Menu / route	CAPA detail · 'Approve' button when status is *_PENDING_APPROVAL
Available to roles	auditor tenant_admin admin superadmin
Backend API	POST /api/capa-v2/capas/:id/approvals

CLICK-BY-CLICK

1. [CLICK](#) Click 'Approve' button
→ Drawer with stage label
2. [CLICK](#) Choose decision (APPROVED / REJECTED / NEEDS_REWORK)
→ required
3. [TYPE](#) Fill comments + (optional) e-signature meaning
→ comments required
4. [CLICK](#) Click 'Submit decision'
→ Status advances to next state per state machine

4.10 Save action plan + add action items

What it does	Capture corrective + preventive + interim actions. Then add granular action-items each with owner + dueDate.
Menu / route	CAPA detail · Action Plan tab
Available to roles	manageRoles
Backend API	PUT /api/capa-v2/capas/:id/action-plan · POST /api/capa-v2/capas/:id/action-items

CLICK-BY-CLICK

1. [CLICK](#) Open Action Plan tab
→ Plan summary form
2. [TYPE](#) Fill correctiveActionsSummary + preventiveActionsSummary
→ AI Draft RCA pre-populates these if invoked from a deviation
3. [CLICK](#) Click '+ Add action item'
→ Item form
4. [TYPE](#) Fill description + assignedToUserId + dueDate
→ All required
5. [CLICK](#) Click 'Submit for approval'
→ Status moves to ACTION_PLAN_PENDING_APPROVAL

4.11 Update action-item status (start / complete / block)

What it does	Action-item owner moves their item through NOT_STARTED → IN_PROGRESS → COMPLETED (or BLOCKED).
Menu / route	Action item card · status dropdown
Available to roles	manageRoles
Backend API	PATCH /api/capa-v2/action-items/:id/status

CLICK-BY-CLICK

1. [CLICK](#) Click status dropdown on an action-item card
→ Status enum shown
2. [CLICK](#) Pick IN_PROGRESS or COMPLETED
→ Item updates; CAPA status moves to IN_IMPLEMENTATION when first item starts

4.12 Upload implementation evidence

What it does	Attach documents proving the action item was completed (test results, updated SOP, training records).
Menu / route	CAPA detail · Implementation tab · '+ Upload evidence' button
Available to roles	manageRoles
Backend API	POST /api/capa-v2/capas/:id/implementation-evidence

CLICK-BY-CLICK

1. [CLICK](#) Click '+ Upload evidence'
→ Drawer with file picker + metadata fields
2. [CLICK](#) Pick file (uploads to S3) or paste link
→ documentRef captured
3. [TYPE](#) Fill description + actionItemId link
→ required
4. [CLICK](#) Click 'Save evidence'
→ evidence row appears under the action item

4.13 Begin + record effectiveness check

What it does	After all actions COMPLETED + grace period passed, run the effectiveness review.
Menu / route	CAPA detail · Effectiveness tab
Available to roles	manageRoles
Backend API	PUT /api/capa-v2/capas/:id/effectiveness

CLICK-BY-CLICK

1. [CLICK](#) Click 'Begin effectiveness review'
→ Status moves to EFFECTIVENESS_REVIEW_IN_PROGRESS
2. [TYPE](#) Fill review window dates + observations
→ Required
3. [CLICK](#) Choose result (PASS / FAIL)
→ Drives close outcome

4.14 Close CAPA

What it does	Final closure. Captures closureOutcome (EFFECTIVE / INEFFECTIVE) + reason. Approval-gated.
Menu / route	CAPA detail · 'Close CAPA' button
Available to roles	approvalRoles
Backend API	POST /api/capa-v2/capas/:id/close

CLICK-BY-CLICK

1. [CLICK](#) Click 'Close CAPA'
→ Confirmation dialog
2. [CLICK](#) Pick outcome (EFFECTIVE / INEFFECTIVE)
→ required
3. [TYPE](#) Fill closureReason
→ required
4. [CLICK](#) Click 'Confirm + sign'
→ Status flips to CLOSED_EFFECTIVE / CLOSED_INEFFECTIVE; metric snapshot written

4.15 Predictive CAPA badge (Wave 3)

What it does	Inline P(on-time) + P(effective) + top factors.
Menu / route	CAPA row badge + drawer card
Available to roles	any viewer
Backend API	POST /api/ai/predict/capa-outcome
AI assistance	AI Heuristic model capa.heuristic@1.0.0 — features: severity, rootCauseDepth, actionCount, daysToDue

CLICK-BY-CLICK

1. WAIT Auto-renders on row open
- Two chips: P(on-time) and P(effective)

5. Full lifecycle walkthrough — one transaction, end-to-end

One CAPA walked from candidate intake through every approval gate to CLOSED_EFFECTIVE. Source = the deviation lifecycle from the Deviation guide.

#	Persona	State transition	Action	API	Observed	Result
1	Kenji QA Specialist	– → INTAKE_DRAFT	From the deviation View+AI drawer, click 'Create CAPA from this RCA' (cross-module). Cross-module endpoint mints a real CAPA + intake.	POST /api/quality/capa-from-deviation	capaNumber=CAPA-2026-NNNN · status=CAPA_OPEN · sourceType=DEVIATION · linked back to deviation.linkedCAPAIds	PASS

Expected DB state after this step:

```
capa-v2 { _id, capaNumber: 'CAPA-2026-NNNN', status: 'CAPA_OPEN', sourceType: 'DEVIATION', sourceRef: { id, collection: 'deviations', label: 'DEV-NNNN' }, ownerUserId, dueDate: +30d, createdBy }
capa-v2-status-history { from: null, to: 'CAPA_OPEN', userId, ts }
capa-v2-action-plans { capaId, correctiveActionsSummary, preventiveActionsSummary }
```

state-screens/capa-buyer.png

Expected DB state after this step:

```
capa-v2-investigations { capaId, scope, summary, findings, evidenceLinks: [] }
capa-v2-status-history { from: 'CAPA_OPEN', to: 'INVESTIGATION_IN_PROGRESS' }
```

#	Persona	State transition	Action	API	Observed	Result
Expected DB state after this step: <pre> capa-v2-root-causes { capaId, rcaMethod: 'FIVE_WHY', rootCauseStatement, verification: {} } capa-v2-status-history { from: 'INVESTIGATION_IN_PROGRESS', to: 'RCA_PENDING_APPROVAL' } </pre>						
4	James Approver (admin)	RCA_PENDING_APPROVAL → ACTION_PLAN_PENDING_APPROVAL	Click 'Approve RCA' → decision=APPROVED → comments → Submit	POST /api/capa-v2/capas/:id/approvals (stage=RCA, decision=APPROVED)	Status advances; capa-v2-approvals row written	PASS
Expected DB state after this step: <pre> capa-v2-approvals { capaId, stage: 'RCA', decision: 'APPROVED', userId, comments, decidedAt } </pre>						
5	Kenji QA Specialist	ACTION_PLAN_PENDING_APPROVAL → ACTION_PLAN_APPROVED	(Already submitted via Create-CAPA; James approves it). Approve action plan.	POST /api/capa-v2/capas/:id/approvals (stage=ACTION_PLAN, decision=APPROVED)	Status moves; one capa-v2-approvals row added	PASS
6	Action-item owners varied	ACTION_PLAN_APPROVED → IN_IMPLEMENTATION	Each owner moves their action-item NOT_STARTED → IN_PROGRESS	PATCH /api/capa-v2/action-items/:id/status	First IN_PROGRESS auto-promotes parent CAPA to IN_IMPLEMENTATION	PASS
Expected DB state after this step: <pre> capa-v2-action-items[*].status='IN_PROGRESS' </pre>						
7	Action-item owners varied	IN_IMPLEMENTATION → AWAITING_EFFECTIVENESS_CHECK	All action-items → COMPLETED + at least one capa-v2-implementation-evidence row uploaded	PATCH /api/capa-v2/action-items/:id/status · POST /api/capa-v2/capas/:id/implementation-evidence	Status moves; evidence rows visible	PASS
Expected DB state after this step: <pre> All capa-v2-action-items.status='COMPLETED' capa-v2-implementation-evidence[*] </pre>						
8	Maria Effectiveness reviewer	AWAITING_EFFECTIVENESS_CHECK → EFFECTIVENESS_REVIEW_IN_PROGRESS	Effectiveness tab → Begin review → reviewWindow + observations → Save	PUT /api/capa-v2/capas/:id/effectiveness	capa-v2-effectiveness-checks row created with result=PENDING	PASS
Expected DB state after this step: <pre> capa-v2-effectiveness-checks { capaId, result: 'PENDING', reviewWindowStart, reviewWindowEnd } </pre>						
9	Elena VP Quality	EFFECTIVENESS_REVIEW_IN_PROGRESS → CLOSED_EFFECTIVE	Click 'Close CAPA' → outcome=EFFECTIVE → closureReason → Confirm	POST /api/capa-v2/capas/:id/close	Status=CLOSED_EFFECTIVE · closedAt set · capa-v2-metric-snapshots row written	PASS
Expected DB state after this step: <pre> capa-v2 { status: 'CLOSED_EFFECTIVE', closedAt, closureOutcome: 'EFFECTIVE' } capa-v2-metric-snapshots { capaId, ageDays, actionItemsClosed, result: 'EFFECTIVE' } </pre>						

6. AI assistance map

Each AI assist is a separate endpoint. Outputs flow through the grounded-generation runtime (citation gate, confidence floor, schema validation, accept/reject audit trail).

AI agent	Attached to state(s)	Endpoint	Where in UI	What it returns	Provider
AI 5-Why Scaffolder (Wave 1)	INVESTIGATION_IN_PROGRESS	POST /api/ai/deviation/scaffold-five-why	RCA tab · 'Scaffold 5-Why with AI' button	5-level Why chain with citations + 6 follow-up questions + 6M categorisation	Free Gemini 2.5 Flash-Lite
AI CAPA RCA Drafter (Wave 1)	INVESTIGATION_IN_PROGRESS → ACTION_PLAN_PENDING_APPROVAL	POST /api/ai/capa/draft-rca	Pre-populates RCA + Action Plan tabs when invoked from a deviation	Full RCA + corrective + preventive + effectiveness draft	Free Gemini 2.5 Flash-Lite
Predictive CAPA Outcome (Wave 3)	any post-CAPA_OPEN	POST /api/ai/predict/capa-outcome	CAPA row badge + detail card	P(on-time) + P(effective) + top contributing factors	Heuristic capa.heuristic@1.0.0
Active Learning Loop	CLOSED_EFFECTIVE CLOSED_INEFFECTIVE	POST /api/ai/active-learning/adjustments	Drift dashboard	Refines AI retrieval-set weights based on accept/reject outcomes	Wave 3 retrieval re-weighting
(roadmap) AI similarity scoring	CAPA_OPEN, CLOSED_*	(not wired)	(future)	Auto-populate capa-v2-similarity-links with SIMILAR/RECURRENCE links from embedding similarity	Vector DB (pgvector planned)

7. Regulator-trace matrix

For each lifecycle state, the specific clauses + evidence Hawkeye captures so an inspector can audit the record. Maps to FDA / ISO / ICH / EU GMP source clauses.

State	Cited clauses	Evidence captured	Backing records
INTAKE_DRAFT	21 CFR 820.100(a)(1) ISO 9001 §10.2.1	issueStatement + issueDescription + sourceType captured	capa-v2-intakes
UNDER_TRIAGE	ICH Q10 §3.2.2	Triage decision rationale + decider + timestamp	capa-v2-triage
INVESTIGATION_IN_PROGRESS	21 CFR 211.192 21 CFR 820.100(a)(2)	Investigation scope + findings + evidenceLinks	capa-v2-investigations
RCA_PENDING_APPROVAL	ICH Q9 §IV 21 CFR 820.100(a)(3)	rcaMethod + rootCauseStatement + verification	capa-v2-root-causes + capa-v2-approvals (stage=RCA)
ACTION_PLAN_APPROVED	21 CFR 820.100(a)(4) ISO 9001 §10.2.1(d)	correctiveActions + preventiveActions + per-item owner + dueDate	capa-v2-action-plans + capa-v2-action-items + capa-v2-approvals (stage=ACTION_PLAN)
IN_IMPLEMENTATION	21 CFR 820.100(a)(4)	Each completed action-item + uploaded evidence (S3 ref + metadata)	capa-v2-action-items + capa-v2-implementation-evidence
EFFECTIVENESS_REVIEW	21 CFR 820.100(a)(4) ICH Q10 §3.2.4	Review window dates + observations + result (PASS / FAIL)	capa-v2-effectiveness-checks
CLOSED_EFFECTIVE	21 CFR 820.100(a)(5) 21 CFR Part 11 §11.50	Closure approval + closedBy + closedAt + closureOutcome + e-sig	capa-v2-approvals (stage=CLOSURE) + capa-v2-status-history (final row)

8. Test results matrix

Suite	Scope	Result	Evidence
eqms-lifecycle.spec.ts · capa-v2	GET workspace · POST intake · POST submit · UI workspace render	PASS	4/4 PASS · eqms-test-results-v2.pdf
eqms-cross-module.spec.ts · F1	Deviation → AI RCA → real CAPA-v2 created in workspace + linked	PASS	F1 steps 3-6 PASS
eqms-cross-module.spec.ts · F3	Complaint → CAPA-v2 intake (sourceType=COMPLAINT) → linked back	PASS	F3 steps 3-4 PASS
Wave 3 smoke	POST /api/ai/predict/capa-outcome	PASS	wave3 smoke

9. Known gaps + roadmap

- **Full UI for the 5 approval stages** — Today the workspace shell is built but the per-tab forms are partially scaffolded. Need full UI for Investigation / RCA / Action Plan / Effectiveness drawers with required-field gating. **HIGH**
- **AI similarity / recurrence detector** — capa-v2-similarity-links table exists but is empty. Plug embedding-based similarity (pgvector) to surface 'this CAPA looks like CAPA-2025-019 — recurrence?' suggestions at intake. **MEDIUM**
- **Action-item dependency graph** — Action-items can have parents but the UI doesn't visualise the DAG. Add a Gantt-style or graph view. **LOW**
- **Effectiveness review reminder scheduler** — When an action plan reviewWindow opens, send a notification to the reviewer (notificationOutbox). **MEDIUM**
- **Bulk-import legacy capas → v2** — Legacy capas collection is read-only. Provide a one-shot migration endpoint that maps old fields to v2 + preserves the audit trail. **LOW**